Bidding-Based Online Policy Adaptation for Evolving Objectives

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Summary

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Contribution(s)

1. Provide a succinct but precise list of the contribution(s) of the paper. Use contextual notes to avoid implications of contributions more significant than intended and to clarify and situate the contribution relative to prior work (see the examples below). If there is no additional context, enter "None". Try to keep each contribution to a single sentence, although multiple sentences are allowed when necessary. If using complete sentences, include punctuation. If using a single sentence fragment, you may omit the concluding period. A single contribution can be sufficient, and there is no limit on the number of contributions. Submissions will be judged mostly on the contributions claimed on their cover pages and the evidence provided to support them. Major contributions should not be claimed in the main text if they do not appear on the cover page. Overclaiming can lead to a submission being rejected, so it is important to have well-scoped contribution statements on the cover page.
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2. The submission template for submissions to RLJ/RLC 2025
   Context: Built from previous RLC/RLJ, ICLR, and TMLR submission templates
3. *[Example of one contribution and corresponding contextual note for the paper "Policy gradient methods for reinforcement learning with function approximation" (**?**).]*
   This paper presents an expression for the policy gradient when using function approximation to represent the action-value function.
   Context: Prior work established expressions for the policy gradient without function approximation (**?**).

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Abstract

We study multi-objective reinforcement learning settings in which objectives appear or disappear at runtime. We propose a modular framework where each objective is supported by a selfish local policy, and coordination is achieved through a novel *auction*-based mechanism: policies bid for the right to execute their actions, with bids reflecting the urgency of the current state. The highest bidder selects the action, enabling a dynamic and interpretable trade-off among objectives. To succeed, each policy must not only optimize its own objective, but also reason about the presence of other goals and learn to produce calibrated bids that reflect relative priority. When objectives change, the system adapts by simply adding or removing the corresponding policies. Moreover, when objectives arise from the same parameterized family—like the class of reachability objectives parameterized by target states—identical copies of a parameterized policy can be deployed. In our implementation, the policies are trained concurrently using proximal policy optimization (PPO). We evaluate on Atari Assault and a gridworld-based path-planning task with dynamic targets. Our method achieves substantially better performance and reduced sample complexity than a single policy trained with PPO.

1 Introduction

A majority of real-world control problems require fulfilling more than one objectives simultaneously, where the objectives could be partly contradictory to each other. We consider the setting where objectives appear or disappear at deployment time, and no prior information about their arrival is available. Our running example is a coffee serving robot in an office building, where new coffee requests could appear from any place at any time, old requests could disappear even before being served, and the robot must react to updated objectives as quickly as possible. In general, any number of requests could be active at any time, and we do not assume any prior knowledge about their distribution. Our goal is to design lightweight, adaptive policies that update their behavior in light of the evolving objectives. Despite having been studied in the planning literature (**?**), this class of problems lack support in (model-free) reinforcement learning (RL), and the existing solutions from the planning setting do not readily extend.

We propose a novel policy adaptation algorithm for this class of problems. The heart of our approach is a compositional design of policies for multi-objective RL problems, where each objective is served using an independent local policy, and coordination is achieved via an online auction-based mechanism: policies bid for the right to execute their actions, with bids reflecting the urgency of the current state for fulfilling the respective objectives. The highest bidder selects the action, enabling a dynamic and interpretable trade-off among objectives. For example, imagine a situation that the coffee serving robot is approaching an intersection of two corridors, there are two active requests and two independent local policies for serving them, the first policy needs to go left, while the second policy needs to go right. Clearly a trade-off is unavoidable, and which of the policies is prioritized is decided based on auctions.

Given this compositional design, every time an objective appears or disappears, only the responsible policy needs to be added or removed, eliminating the need for modifying the whole system. Moreover, when the objectives come from the same parameterized family, like the set of all possible coffee requests parameterized by the request locations, we can design a single universal policy for this entire parameterized family, so that new additions require adding an identical copy of the policy, offer us instant adaptation. In contrast, a monolithic policy that would serve all objectives lacks this flexibility, and an attempt to train such a policy with variable number of objectives causes a dramatic degradation of training stability as well as the performance (the loss); we demonstrate this in our experiments.

We show how classical RL policies can be extended with bidding capabilities. While the obvious first step is to extend the action space with numeric bid values, there are three key challenges:

Challenge I: enforcing honest bids. We must ensure that policies bid only in proportion to their urgency, because otherwise, there is a risk of obtaining "dishonest" policies that constantly try to block others by overbidding. To circumvent this issue, during training, we require the policies to pay penalties proportional to their bid values. **KM:** I suggest, we just stick to one kind of bidding, instead of three, if in the end all-pay remains superior in all experiments. This way, it is against the interest of policies to bid too high, because otherwise, the net benefit of achieving their objectives would be lost.

Challenge II: achieving environment awareness. The bidding tactic of policies should not only depend on the current state and the own objective, but also needs to account for other objectives for maximal effectiveness. For instance, if two coffee requests appear at nearby locations, the respective policies need not bid too high to compete against the opponent. Dually, if two requests arrive from opposite directions, the bid of the more urgent policy must be high enough to counteract the opponent. Therefore, the policies must account for the objectives of opponents to bid effectively. Our approach is to model the local policy synthesis problem as an instance of multi-agent nonzerosum games, where each agent is responsible for an individual objective, and needs to learn a policy that fulfills its objective against the opponents.

Challenge III: unbounded objective count. Since the number of objectives seen during deployment cannot be predicted during training, the game-theoretic approach for environment-aware policies falls apart, because now we face a variable number of opponents for which no solution is known. We use an attention network that transforms an arbitrary number of opponent objectives to a fixed encoding, which is then fed as the input to the RL policy. Each local policy is equipped with its own attention network, which is co-trained with the actual policy for an increasing number of opponent objectives. **KM:** If we want to make the unbounded objectives a central feature, we should restrict ourselves to objectives from the same family. Otherwise, the attention pooling and the game-theoretic training approach does not work. While this is indeed a restrictive case, it is well-motivated and I do not see it as a weakness.

We implemented our framework using the proximal policy optimization (PPO) as the base learning algorithm. Using the Atari Assault and a gridworld-based path-planning task, we demonstrate the superior training stability and the performance of our policies in comparison with the baseline monolithic policies. **TODO:** Summarize the main take-away: can we give some statistics, like "the training time was 2x faster" or the "loss was 2x smaller"?

1.1 Related work

A large body of work in multi-objective reinforcement learning (MORL) relies on *scalarization*, aggregating multiple reward functions into a single scalar objective so that standard single-objective RL algorithms can be applied. The simplest scalarization method is a weighted sum of individual rewards (Gass & Saaty, 1955), though richer nonlinear scalarization functions have also been proposed (Van Moffaert et al., 2013). A key limitation of scalarization is that the relative importance induced by the aggregation function may not align with the designer's true intent. This mismatch

can initiate a tedious debugging cycle, particularly in large-scale systems (Hayes et al., 2022). In contrast, our approach achieves a trade-off between reward components without collapsing them into a fixed scalar objective.

Other works pursue trade-offs by fixing a specific optimality criterion. Common choices include Pareto optimality (Van Moffaert & Nowé, 2014) and its approximations (Pirotta et al., 2015), as well as fairness-based criteria across reward functions (Park et al., 2024; Byeon et al., 2025; Siddique et al., 2020). These approaches typically learn a single monolithic policy that satisfies the chosen criterion. By contrast, our objective is to learn independent, selfish local policies for each reward component and compose them at runtime in a principled manner, thereby preserving modularity while still achieving a coherent global trade-off.

Relatively few works study distributed local policies for multiple rewards. A notable example is W-learning (Humphrys, 1995) and its deep RL extension (Rosero et al., 2024), where separate selfish policies are trained alongside meta-policies (W-functions) that assign each state a score reflecting its urgency. At runtime, the policy with the highest score is selected. Other approaches employ alternative aggregation mechanisms, such as ranked voting over actions (Méndez-Hernández et al., 2019), or fixed aggregation rules like summing action values across agents (Russell & Zimdars, 2003). While conceptually related, our approach is technically simpler: it relies on an engineered reward structure that enables the use of standard learning algorithms (e.g., PPO) without additional meta-policies or complex aggregation schemes. Furthermore, to the best of our knowledge, we are the first to introduce the incremental MORL setting, in which reward components can be added or removed at runtime.

The idea of bidding-based selfish policies originates from analogous techniques for multi-objective path planning problems on finite graphs (Avni et al., 2024), as well as from the broader literature on bidding games (Lazarus et al., 1999; Avni et al., 2019; 2025). These works study strategic interaction in finite arenas, where adversarial players bid for the right to determine the next move from a shared action space in pursuit of their objectives. Although these works provide strong theoretical guarantees, they do not naturally extend to infinite arenas. Moreover, players in such games are typically budget-constrained, and the central question concerns the minimum budget required to win. In contrast, we consider infinite arenas and eliminate explicit budget constraints by incorporating bidding rewards and penalties directly into the learning framework.

2 Preliminaries: Multi-Objective MDPs

Mainstream RL algorithms consider Markov decision processes (MDP) equipped with a *single* reward function, pertaining to a single task or *objective* for the system. In reality, a majority of real-world applications of RL requires satisfying multiple, partly contradictory objectives. We model such multi-objective decision-making problems using multi-objective MDPs (MO-MDP), as formally defined below. Intuitively, an MO-MDP has the exact same syntax as a regular MDP, except that it now has multiple reward functions pertaining to the different objectives. We formalize MO-MDP below. We will use the notation $\mathbb{D}(\Sigma)$ to represent the set of all probability distributions over a given alphabet Σ.

Definition 1 (MO-MDP)**.** A multi-objective Markov decision process (MO-MDP) with $m \in \mathbb{Z}_{>0}$ objectives is specified by a tuple $\mathcal{M} = (S, A, T, \mathbf{R}, \mu_0)$, where

- S is the set of states,
- A is the set of actions,
- $T : S \times A \to \mathbb{D}(S)$ is the transition function mapping a state-action pair to a distribution over the successor states,
- $\mathbf{R} = \{R^i : S \times A \times S \to \mathbb{R}_{\geq 0}\}_{i \in [1;m]}$ is the set of reward functions, and
- $\mu_0 \in \mathbb{D}(S)$ is the initial state distribution.

The notions of policies and paths induced by them are exactly the same as in classical MDPs, which we briefly recall below. First, we introduce some notation. Given an alphabet Σ, we will write Σ^* and Σ^ω to denote the set of every finite and infinite word over Σ, respectively, and will write $\Sigma^\infty = \Sigma^* \cup \Sigma^\omega$. Given a word $w = \sigma_0\sigma_1 \ldots \in \Sigma^\infty$, and given a $t \geq 0$ that is not larger than the length of w, we will write w_t and $w_{0:t}$ to denote respectively the t-th element of w, i.e., $w_t = \sigma_t$, and the prefix of w up to the t-th element, i.e., $w_{0:t} = \sigma_0 \ldots \sigma_t$.

A *policy* in an MO-MDP $\mathcal{M}$ is a function $\pi : (S \times A)^* \times S \to \mathbb{D}(A)$ that maps a history of state-action pairs and the current state to a distribution over actions. A *path* on $\mathcal{M}$ induced by π is a sequence $\rho = (s_0, a_0)(s_1, a_1), \ldots \in (S \times A)^\infty$ such that for every $t \geq 0$, (1) the probability that the action a_{t+1} is picked by π based on the history is positive, i.e., $\pi(\rho_{0:t}, s_{t+1})(a_{t+1}) > 0$, and (2) the probability of moving to the state s_{t+1} from s_t due to action a_t is positive, i.e., $T(s_t, a_t)(s_{t+1}) > 0$. A path can be either finite or infinite, and we will write $Paths(\mathcal{M}, \pi)$ to denote the set of all infinite paths fo $\mathcal{M}$ induced by π. Given a finite path $\rho = (s_0, a_0) \ldots (s_t, a_t)$, the probability that ρ occurs is given by: $\mu_0(s_0) \cdot \prod_{k=0}^{t-1} T(s_k, a_k)(s_{k+1}) \cdot \pi(\rho_{0:k}, s_{k+1})(a_{k+1})$. This can be extended to a probability measure over the set of all infinite paths in $\mathcal{M}$ using standard constructions, which can be found in the literature (Baier & Katoen, 2008). Given a measurable set of paths Ω and a function $f \colon Paths(\mathcal{M}, \pi) \to \mathbb{R}$, we will write $\mathbb{P}^{\mathcal{M},\pi}[\Omega]$ and $\mathbb{E}^{\mathcal{M},\pi}[f]$ to denote, respectively, the probability measure of Ω and the expected value of f evaluated over random infinite paths.

We will use the standard discounted reward objectives, where we fix $\gamma \in (0, 1)$ as a given discounting factor. Let $\rho = (s_0, a_0)(s_1, a_1), \ldots \in Paths(\mathcal{M}, \pi)$ be an infinite path induced by π. Define the discounted sum function, mapping ρ to the discounted sum of the associated rewards: $f^i_{\text{ds}}(\rho) \coloneqq \sum_{t=0}^{\infty} \gamma^t \cdot R^i(s_t, a_t)$. The *$i$-value* of the policy ρ for $\mathcal{M}$ is the expected value of the discounted sum of the i-th reward we can secure by executing ρ on $\mathcal{M}$, written as $val^{\mathcal{M},i}(\pi) = \mathbb{E}^{\mathcal{M},\pi}[f^i_{\text{ds}}]$. The *optimal* policy for R^i for a given $i \in [1; m]$ is the policy that maximizes the i-value. When the reward index i is unimportant, we will refer to every element of the set $\{val^{\mathcal{M},i}\}_{i \in [1;m]}$ as a *value component*.

When the MO-MDP $\mathcal{M}$ is clear from the context, we will drop it from all notation and will simply write $Paths(\pi)$, $\mathbb{P}^\pi$, $\mathbb{E}^\pi$, and val^i.

It is known that *memoryless* (aka, stationary) policies suffice for maximizing single discounted reward objectives, where a policy π is called memoryless if the proposed action only depend on the current state. In other words, given every pair of finite paths ρ, ρ' both ending at the same state, the probability distributions $\pi(\rho)$ and $\pi(\rho')$ are identical.

Unlike classical single-objective MDPs, the optimal policy synthesis problem for MO-MDP requires fixing one of many possible optimality criteria. Many possibilities exist, including pareto optimality, requiring a solution where none of the value components could be unanimously improved without hurting the others; weighted social welfare, requiring a weighted sum of the value components be maximized; and fairness, requiring the minimum attained value by any value component is maximized. **TODO:** Give some citations for each category.

3 Auction-Based Compositional RL on Multi-Objective MDPs

We consider the compositional approach to policy synthesis for MO-MDPs, where we will design a selfish, *local* policy maximizing each individual value component, towards the fulfillment of some required global coordination requirements. The main crux is in the composition process, where each local policy may propose a different action, but the composition must decide one of the actions that will be actually executed. Importantly, the composition must be implementable in a distributed manner, meaning we will *not* use any global policy that would pick an action by analyzing all local policies and their reward functions. **TODO:** running example

3.1 The Framework

We present a novel *auction*-based RL framework for compositional policy synthesis for MO-MDPs. In our framework, not only do the local policies emit actions, but also they *bid* for the privilege of executing their actions for a given number of time steps $\tau \in \mathbb{N}_{>0}$ in future. The bids are all non-negative real numbers, and the highest bidder's actions get executed for the following τ consecutive steps, with bidding ties being resolved uniformly at random. The policy whose actions are executed is referred to as the *winning* policy, and it must pay a bidding *penalty* that equals to its bid amount; this is to discourage overbidding. The policies whose actions are not executed are called the *losing* policies, and we consider three different settings for the "payment" they must make:

Loser-Rewarded: the winning policy pays the bidding penalty and the losing policies earn bidding rewards equal to their respective bid values;

Winner-Pays: the winning policy pays the bidding penalty and the losing policies are unaffected (i.e., neither earn bidding rewards nor pay bidding penalties);

All-Pay: all policies pay bidding penalties equal to their respective bid values.

While penalizing the winner discourages overbidding, the situation with the losers is more subtle. In the Loser-Rewarded setting, by rewarding the losers, we encourage policies to bid positively if the current state has some importance to them; this way, if they lose the bidding, they will get some positive reward. In the All-Pay setting, by penalizing all policies, we discourage policies to bid at all unless it is absolutely important. The Winner-Pays setting balances these two: by neither rewarding nor penalizing the losers, we neither encourage nor discourage policies to bid. In Section 3.3, we will see how these three settings induce different kinds of coordination through bidding.

For each policy, the bidding penalty or reward gets, respectively, subtracted or added to the *nominal* reward obtained from the reward functions of the given MO-MDP, and the resulting reward is called the *net* reward.

In summary, through this novel bidding mechanism, each policy can adjust its bid in proportion to the importance for it to execute its action in the current state, and the associated bidding penalty/reward aims to incentivize policies to be truthful. By making the highest bidder active, it is effectively guaranteed that the most important policy is executed. This way, we obtain a purely decentralized scheme to coordinate local policies in a given MO-MDP.

Remark 1 (On the parameter τ). The parameter τ controls how frequently the agent changes its policies. In practice, if τ is too small, the switching could be too frequent for any of the objectives to be fulfilled. For example, **TODO:** running example...

3.2 The Design Problem and Learning Algorithms

We consider the following learning task for our auction-based compositional framework:

Given an MO-MDP, a constant $\tau > 0$, and $\Delta \in \{$Loser-Rewarded, Winner-Pays, All-Pay$\}$, compute local policies that are optimal for the net rewards obtained in the mode Δ, given that all other local policies behave selfishly towards maximizing their own net rewards.

We will show how the above learning problem boils down to solving a standard learning problem in the multi-agent setting, formalized using a decentralized MDP (DEC-MDP) as defined below. The only difference between a DEC-MDP and an MO-MDP (see Definition 1) is that now each reward function R^i is owned by the Agent i, who now controls a separate set of actions A^i.

Definition 2 (DEC-MDP)**.** A decentralized Markov decision process (DEC-MDP) with $m \in \mathbb{Z}_{>0}$ agents is specified by a tuple $\mathcal{M} = (S, \mathbf{A}, T, \mathbf{R}, \mu_0)$, where

- S is the set of states,
- $\mathbf{A} = \{A^1, \ldots, A^m\}$ is a set with A^i being the set of Agent i's actions,

- $T : S \times A^1 \times \ldots \times A^m \to \mathbb{D}(S)$ is the transition function mapping a state-action pair to a distribution over the successor states,
- $\mathbf{R} = \{R^i \colon S \times A^1 \times \ldots \times A^m \times S \to \mathbb{R}_{\geq 0}\}_{i \in [1;m]}$ is the set of reward functions, and
- $\mu_0 \in \mathbb{D}(S)$ is the initial state distribution.

The definitions of policies and paths readily extend from MO-MDP to DEC-MDP.

Given a DEC-MDP, the goal is to compute an ensemble of local (memoryless) policies for all individual agents, such that for every $i \in [1;m]$, the i-value cannot be increased by a unanimous change of the local policy π^i. In other words, the goal is to find a set of selfish local policies that are in a Nash equilibrium. This is an extensively studied problem in the literature. **TODO:** Do a little bit of literature survey...

Our focus is not in improved algorithms for DEC-MDP, but rather to show how the local policy synthesis problem for the MO-MDP $\mathcal{M}$ in our auction-based framework reduces to the multi-agent policy synthesis problem in a DEC-MDP $\widetilde{\mathcal{M}}$. Intuitively, for every state s of $\mathcal{M}$, $\widetilde{\mathcal{M}}$ creates two kinds of copies, ones where bidding happens and are represented simply as s, and ones of the form (s, t, i^*) that keeps track of the time t elapsed since the last bidding, and the winner i^* of the last bidding. Furthermore, bidding is facilitated by extending the action space of $\mathcal{M}$ to include all real-valued bids, and each agent in $\widetilde{\mathcal{M}}$ has an identical copy of this extended action space. After bidding in a state s, the winner i^* is selected, and the state moves to $(s, 0, i^*)$. From this point onward, only Agent i^* selects actions $a^0, a^1, \ldots, a^\tau$ to produce the sequence $(s^1, 1, i^*), (s^2, 2, i^*), \ldots, (s^{\tau-1}, \tau - 1, i^*), s^\tau$, after which the next bidding happens, and the process repeats. Finally, the bidding penalties or bidding rewards are only paid during the transition $s \to (s, 0, i^*)$, otherwise, the rewards are inherited from the original MO-MDP.

We formalize this below. Given an MO-MDP $\mathcal{M} = (S, A, T, \mathbf{R}, \mu_0)$, a constant $\tau > 0$, and the mode $\Delta \in \{\textsf{Loser-Rewarded}, \textsf{Winner-Pays}, \textsf{All-Pay}\}$, we define the DEC-MDP $\widetilde{\mathcal{M}} = (\widetilde{S}, \widetilde{\mathbf{A}}, \widetilde{T}, \widetilde{\mathbf{R}}, \widetilde{\mu}_0)$ where

- $\widetilde{S} \coloneqq S \cup S \times [0; \tau - 1] \times [1; m]$,
- $\widetilde{\mathbf{A}} \coloneqq \{\widetilde{A}^i\}_{i \in [1;m]}$ where $\widetilde{A}^i \coloneqq A \cup \mathbb{R}$,
- $\widetilde{\mu}_0 \coloneqq \mu_0 \times \{0\}$,

and for every current state $s \in \widetilde{S}$ and every current action $(b^1, \ldots, b^m) \in \mathbb{R}^m$, writing the highest bidders as $I = \{i \in [1;m] \mid \forall j \in [1;m] \,.\, b^i \geq b^j\}$,

- $\widetilde{T}(s, b^1, \ldots, b^m) \coloneqq \text{Uniform}(\{(s, 0, i)\}_{i \in I})$,
- $\widetilde{R}^i(s, b^1, \ldots, b^m, (s, 0, i^*)) \coloneqq \begin{cases} -b^i & i = i^* \vee \Delta = \textsf{All-Pay}, \\ +b^i & i \neq i^* \wedge \Delta = \textsf{Loser-Rewarded}, \\ 0 & i \neq i^* \wedge \Delta = \textsf{Winner-Pays}, \end{cases}$

whereas if the current state is of the form $(s, t, i^*) \in \widetilde{S}$, for every action $(a^1, \ldots, a^m) \in A^m$,

- $\widetilde{T}((s, t, i^*), a^1, \ldots, a^m) \coloneqq \begin{cases} T(s, a^{i^*}) \times ((t+1) \mod \tau) \times \{i^*\} & t < \tau - 1, \\ T(s, a^{i^*}) & t = \tau - 1, \end{cases}$
- $\widetilde{R}^i((s, t, i^*), a^1, \ldots, a^m, (s', t+1, i^*)) \coloneqq R^i(s, a^i, s')$.

KM: A soundness theorem would be good, but what can we say concretely?

3.3 Flavors of Cooperation through Bidding

We provide theoretical insights into the global behavior that emerges out of the auction-based interactions between the local policies. For the sake of theoretical guarantees, and to be able to convey the main essence of our results, we choose the simplest bare bone setting:

Assumption 1. The given MO-MDP has finite state and action spaces, and for every (memoryless) policy, the bottom strongly connected component (BSCC) of the resulting Markov chain (MC) is a sink state where no reward is earned. Furthermore, the time parameter $\tau = 1$, meaning the bidding takes place at each time step before selecting the action.

Firstly, since the MO-MDP is finite, for each individual reward function, *deterministic* memoryless policy suffices. **TODO:** give some citation

The following two types of global behaviors are of particular interest:

Social welfare is the sum (equivalently, the average) of the i-values for all i. We may ask: is the emergent global behavior guaranteed to achieve the maximal social welfare?

Fairness is measured by the disparity between different i-values, i.e., $\max_{i,j\in[1;m]} |val^i - val^j|$. Fairness is maximized when the disparity is minimized. We may ask: is the emergent global behavior guaranteed to achieve the maximal fairness?

Theorem 1. *Suppose the MO-MDP is such that at each state s and for every action a, there exists at most a single $i \in [1;m]$ such that the optimal policy for R^i selects a at s. Then, the* Loser-Rewarded *setting maximizes the social welfare.*

Proof sketch. First, consider the simple one-shot game, where the agents bid just one time to select an action, and the reward is based on the resulting single probabilistic transition. Suppose for the index $i \in [1;m]$, the expected reward from using the action $a \in A$ is E^i_a, and define $E^i_+ := \max_{a\in A} E^i_a$ and $E^i_- := \min_{a\in A} E^i_a$.

We claim that the optimal bid b^i_* for policy i equals $(E^i_+ - E^i_-)/2$, and upon winning the bidding the optimal action is $a_+ = \arg\max_{a\in A} E^i_a$. Notice that no matter whether policy i becomes the winner or the loser, its net reward is at least $(E^i_+ + E^i_-)/2$: if it wins and chooses a_+, after paying the bidding penalty, the net reward is $E^i_+ - (E^i_+ - E^i_-)/2 = (E^i_+ + E^i_-)/2$; if it loses, no matter what action the opponent chooses, its nominal reward is at least E^i_-, and after the bidding reward, the net reward is $E^i_- + (E^i_+ - E^i_-)/2 = (E^i_+ + E^i_-)/2$. If policy i bids $b^i < b^i_*$, then upon losing, its net reward will be $E^i_- + b^i < E^i_- + b^i_* = (E^i_+ + E^i_-)/2$. If it bids $b^i > b^i_*$, then upon winning, its net reward will be $E^i_+ - b^i < E^i_+ - b^i_* = (E^i_+ + E^i_-)/2$. Therefore, the optimal bid is $b^i_* = (E^i_+ - E^i_-)/2$, which is what each selfish policy is expected to select.

Suppose, policy i is the winner. Then, for every $j \neq i$, $b^i_* \geq b^j_*$, i.e., $(E^i_+ - E^i_-)/2 \geq (E^j_+ - E^j_-)/2$. Simplifying, we get $E^i_+ + E^j_- \geq E^i_- + E^j_+$. It follows that $E^i_+ + \sum_{j\neq i} E^j \geq E^i_+ + \sum_{j\neq i} E^j_- \geq E^i_- + E^k_+ + \sum_{j\neq i,k} E^j$ for every for every $k \neq i$. Since the MO-MDP is purely competitive, there will be at least a single k such that a given action is optimal for k, and therefore the claim follows for the single-shot case.

Now, for the general multi-shot case, we inductively apply the above principle in the Bellman equation, which extends the claim to paths of arbitrary length. The convergence of the Bellman iteration is guaranteed because it is a contraction mapping (since $\gamma < 1$). **KM:** I am not sure about this extension. □

4 Multi-Policy Bidding for Multi-Objective RL

The base environment model that we consider is described by a multi-objective Markov decision process (MO-MDP). An MO-MDP with $m \in \mathbb{N}_{>0}$ objectives is specified by a tuple $\mathcal{M} = (S, A, T, R, \mu_0)$ where S is the set of states, A is the set of actions, $T : S \times A \to \mathbb{D}(S)$ is the

transition function mapping a state-action pair to a distribution over states, $\mathbf{R} = \{R^i \colon S \times A^1 \times \ldots \times A^m \times S \to \mathbb{R}_{\geq 0}\}_{i \in [1;m]}$ is the set of reward functions corresponding to the m objectives, and finally $\mu_0 \in \mathbb{D}(S)$ is the initial state distribution. A *policy* in an MO-MDP $\mathcal{M}$ is a function $\pi : (S \times A)^* \times S \to \mathbb{D}(A)$ that maps a history of state-action pairs and the current state to a distribution over actions. A policy naturally induces a distribution over sequences of state-action pairs that we denote $\pi_{\mathcal{M}}$.

In this work, we focus on maximizing the sum of rewards across the objectives. Concretely, we seek to find $\arg\max_\pi J(\pi)$ where

$$J(\pi) = \mathbb{E}_{(s_i,a_i)\sim\pi_{\mathcal{M}}} \left[\sum_{t=0}^{\infty} \gamma^t \sum_{i=1}^{m} R^i(s_i, a_i, s_{i+1}) \right] \tag{1}$$

and $\gamma \in (0, 1)$ is the discount factor.

5 Environments with Evolving Objectives

We begin by describing the multi-objective environments that we evaluate our framework on. Despite their practical relevance, environments with multiple evolving objectives have received little attention in the multi-objective RL literature.

Dynamic Targets. We implement a simple but challenging gridworld environment called Dynamic Targets which consists of an $n \times n$ grid with m targets that are moving at a small rate and stochastically change directions. Each target stays alive for a certain number of timesteps before which the agent has to reach it. Once the target expires or the agent reaches it, another new target immediately respawns at a random position in the grid. The metric the agent has to maximize is the difference between the number of targets reached and the number of targets that expired. This flexible environment also allows us to introduce more targets at deployment time. **TODO:** Reference image for benchmark

In our experiments, we choose a 30×30 grid with 8 targets. The observations for all the local policies includes information about all the targets (positions and expiry times) and the agent; but they are reordered so that the specific target of a policy is present at the first index. This enables sharing the same actor network across policies. When the agent reaches a target before the expiration, we reward the corresponding policy and penalize it if not. Both the reward and the penalty are equal in magnitude. Furthermore, we include a distance based reward shaping for each policy with respect to its target which is crucial to the learning. **TODO:** Include this as an advantage of our framework in introduction.

Atari Assault. We also test our method on the Atari game Assault. Here, we control a missile and shoot alien ships that move. Even though this is not a multi-objective environment, we decompose it so that we can treat each alien ship as a different objective. Since these alien ships move around and there are a variable number of ships at any point of time, it is a perfect testbed for our modular approach. We check whether we can synthesize interpretable policies that split control of the missile to target different ships. However, as the standard Atari games from the Arcade Learning Environment (Bellemare et al., 2013) either use image or raw RAM observations, we cannot decompose the game score and design rewards for the multiple policies. Instead, we turn to the Object-Centric Atari environments (Delfosse et al., 2023) which exposes object positions and states by decoding RAM states. This permits us to design decomposed observations and rewards for each policy to pursue and shoot the alien ships. However, the metric we optimize is still the game score.

The observation space consists of the positions of the cannon, the alien ships and enemy missiles, current health level, and the number of remaining lives. Similar to Dynamic Targets, we reorder the positions of the alien ships in the observations for each policy. Instead of rewards based on the game score, we design the rewards for each policy. We reward a policy if it successfully destroys an alien ship and penalize it if it loses a life. **TODO:** Add image of Assault.

Table 1: Performance (mean with 95% CI) averaged over the last 5 evaluation checkpoints.

Algorithm	Gridworld (Min Targets Reached)	Assault (Score)
All-Pay	6.05 [5.74, 6.36]	634.80 [591.14, 678.46]
Winner-Pays	4.07 [3.78, 4.36]	578.04 [521.02, 635.06]
Winner-Pays (Others Rewarded)	4.20 [3.92, 4.48]	662.60 [619.09, 706.11]
Single-Agent	2.31 [2.08, 2.54]	384.72 [343.09, 426.35]
DWN	0.98 [0.75, 1.21]	256.20 [205.12, 307.28]

6 Implementation

Yu et al. (2022) demonstrate that proximal policy optimization (PPO; Schulman et al. (2017)) can be effective to train multiple policies concurrently for multi-agent settings. We follow their recipe and train all the local policies with PPO. Similar to Yu et al. (2022), since the objectives in our environments share the same characteristics, we use the same actor and critic network for all the policies. At each iteration of PPO, we collect all the rollouts from each policy's perspective and sample minibatches from this pool to perform the PPO updates.

Additionally, in the Dynamic Targets environments, we employ an attention pooling architecture to encode information about all the targets. This allows each local policy to access information about all targets being pursued by competing policies. It also enables competition with an arbitrary number of policies (i.e., more targets) at deployment time. Each target information vector is passed through an MLP to obtain a target embedding. We then take a weighted sum of these embedding vectors where the weights are obtained by taking the dot product of the embedding vector with a learned query vector. This weighted sum is accessible to all the policies at each step.**TODO:** Talk more about action windows maybe

6.1 Baselines

We compare our method against two baselines: a single policy PPO with weighted rewards and the deep W-learning (DWN; Rosero et al. (2024)). For the PPO baseline, since all the objectives have the same importance across the two environments, we use the total reward across objectives at each step as the reward function. The observation space remains the same as the multi-policy setting. However, we do not implement any reward shaping for the Dynamic Targets environment due to the difficulty of specifying a meaningful dense reward signal in the presence of multiple targets. DWN is another multi-policy method for multi-objective RL that performs DQN for each of the local policies and additionally learns a W-network for each policy that predict scalar weights. These weights indicate which policy's action should be played at each step. The W-network is trained alongside the Q-network by updating it with a gradient step to match the negative TD-error since this indicates how important it is for the policy to play its action. The implementation is borrowed from their repository.

7 Experimental results

We conduct experiments to show the superior performance of our method against the baselines. We also perform ablations with respect to all the key components of the framework and draw insights on how we can tune them.

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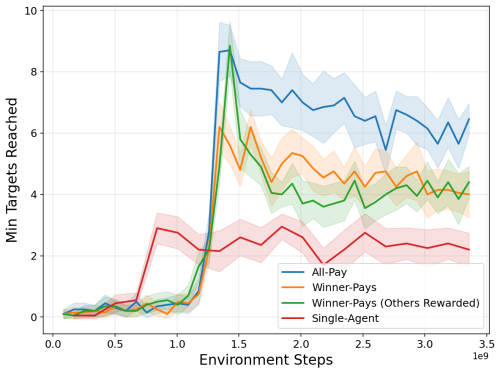
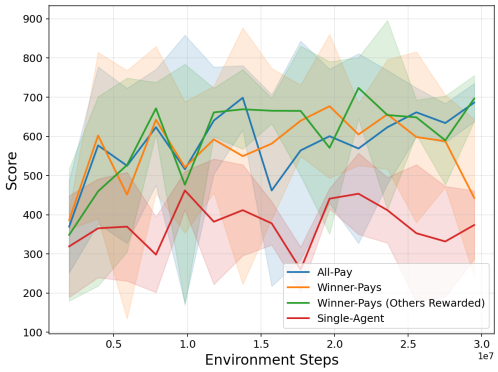
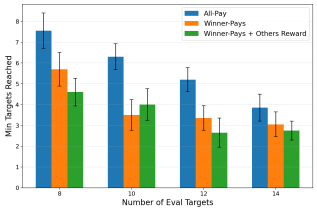


(a) MovingTargetsGridworld. [Placeholder: describe convergence behavior, relative performance of mechanisms, and any notable differences in sample efficiency.]

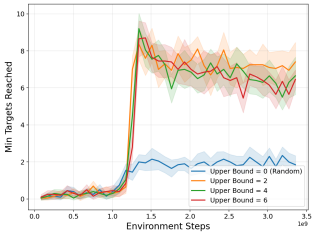


(b) Atari Assault. [Placeholder: describe convergence behavior, relative performance of mechanisms, and any notable differences in sample efficiency.]

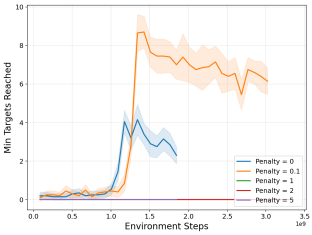
Figure 1: Learning curves for different bidding mechanisms across both environments.



(a) Bidding mechanism scaling. [Placeholder description.]



(b) Bid upper bound. [Placeholder description.]



(c) Bid penalty. [Placeholder description.]

Figure 2: Ablation studies on the MovingTargetsGridworld environment.

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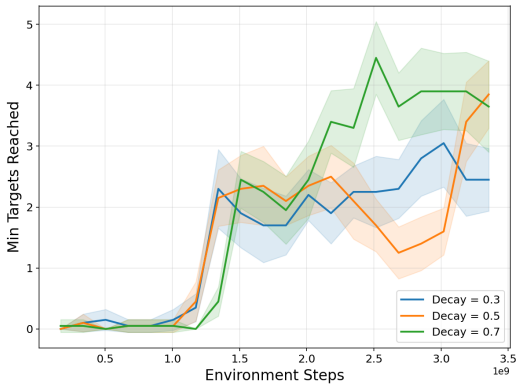

Figure 3: Reward decay. [Placeholder description.]

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